Prof. Dr. Ulrich Bunke

Universität Regensburg

WS 2022

Noncommutative Homotopy theory
Exercise 1

to be discussed: 12.10.2026

Exercise 1. Spectral radius is bounded by the norm: Let $(A, \|-\|)$ be a Banach algebra (Banach space with algebra structure such that $\|ab\| \leq \|a\|\|b\|$) and consider a in A and λ in $\mathbb{C}$. Show that if $|\lambda| > \|a\|$, then $a - \lambda$ is invertible in A. Hint: Neumann series.

Exercise 2. C^*-norm properties: Let H be a Hilbert space and $B(H)$ be the Banach algebra of bounded operators on H. Show that for a in $B(H)$ we have $\|a^*\| = \|a\|$ and $\|a^*a\| = \|a\|^2$. Hint: $\|h\| = \sup_{\|h'\|=1} |\langle h', h\rangle|$.

Exercise 3. Categorical characterization of exact sequences: We work in the category **ban** and consider a square of morphisms.

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & C \end{array} \quad .$$

Show that the following assertions are equivalent:

1. The sequence $0 \to A \to B \to C \to 0$ is exact.
2. The square commutes and is cartesian and cocartesian.

Exercise 4. Completion of Banach spaces as adjunction: We consider the categories **norm** of semi-normed vector spaces (pairs $(V, \|-\|)$ where $\|-\|$ is like a norm but without the axiom that $\|x\| = 0$ implies $x = 0$) and bounded maps and its full subcategory **ban** of Banach spaces over $\mathbb{C}$. Show that there exists an adjunction

$$\mathrm{compl} : \mathbf{norm} \leftrightarrows \mathbf{ban} : \mathrm{incl} \ .$$

Describe the completion functor compl explicitly.